

Energy-Aware Target Streams for Learned Fixed-Wing Maneuvering

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Abstract: Learned aerial control has progressed rapidly on multirotors, but learned fixed-wing maneuvering remains challenging because fixed-wing aircraft must generate maneuvers through coupled attitude, lift, velocity direction, energy, and actuator-limited dynamics rather than hover-like position regulation. This paper studies fixed-wing maneuvering as an executable skill-composition problem. We propose *energy-aware target streams*, a framework that trains a direct-actuator attitude–speed skill, expands its vertical maneuvering envelope with energy- and safety-aware PPO fine-tuning, and composes long-horizon maneuvers by selecting low-dimensional heading–pitch–roll–airspeed target streams through closed-loop rollout. The approach deliberately optimizes executable target streams rather than open-loop actuator sequences or purely geometric waypoint commands. We first show that quaternion conditioning provides a more reliable reusable executor than Euler-angle targets on complex curve-following tasks, while geometry-aware metrics reveal *pseudo-tracking*: low cross-track error can still coincide with poor tangent alignment, unsafe aerodynamic state, energy loss, or incorrect wing-plane orientation. Across S-curve, figure-eight, helix, and 90° vertical pull-up tasks, closed-loop stream selection improves CTE-P90 by 39%, 2%, 24%, and 47%, respectively. The energy-aware executor further completes vertical-arc and full-loop-like maneuvers in simulation. Compared with a classical PID+nonlinear-guidance cascade on a 90° pull-up, the learned executor accepts slightly larger geometric error but avoids overload and completes the climb, illustrating that fixed-wing maneuvering requires dynamics-aware execution rather than CTE minimization alone. The results suggest that the main remaining challenges shift from basic maneuver feasibility to execution quality, energy margins, and sim-to-real validation.

Code: anonymous.4open.science/r/Energy-Aware-Target-Streams-B550

Keywords: fixed-wing aerial robots, reinforcement learning, maneuver composition, target streams, closed-loop stream selection

1 Introduction

Learning-based aerial control has advanced rapidly on multirotor platforms, but learned fixed-wing maneuvering remains comparatively underexplored. The gap is not only one of platform choice: fixed-wing aircraft cannot hover, stop at a waypoint, or rotate in place. Their maneuvers are governed by coupled attitude, lift, velocity direction, energy, actuator authority, and safety margins. As a result, cross-track error (CTE) alone is insufficient: a trajectory may stay close to a spatial reference while the aircraft loses energy, drifts out of the maneuver plane, or flies with inconsistent nose and velocity directions.

This paper studies learned fixed-wing maneuvering as an executable skill-composition problem. Instead of placing learning above a hand-tuned PID/autopilot inner loop, we train a direct-actuator

reinforcement-learning skill that outputs throttle, elevator, aileron, rudder, and speed brake. This exposes the learned policy to the actuator limits, aerodynamic coupling, and energy loss that determine whether a maneuver command is actually executable.

Our method follows a failure-driven progression. An Euler-angle direct-RL baseline remains useful on simple targets, but degrades on complex turning and curve-following tasks. We therefore use a quaternion-conditioned attitude–speed interface as the reusable low-level skill. Geometry-aware evaluation then reveals pseudo-tracking: low CTE can coexist with poor tangent alignment, wing-plane inconsistency, unsafe angle of attack, or energy loss. To address this failure mode, we introduce energy-aware PPO fine-tuning over vertical-arc curricula, safety/energy rewards, and replay.

For long-horizon maneuvers, we do not optimize actuator sequences directly. Instead, we convert each reference maneuver into an executable stream of local heading, pitch, roll, and airspeed targets. We then select low-dimensional stream parameters by closed-loop rollout through the frozen learned skill and fixed-wing dynamics, so the chosen stream is evaluated by actual execution rather than open-loop geometric closeness.

In summary, this paper contributes: (i) a direct-actuator fixed-wing RL skill with an Euler-to-quaternion target evolution, (ii) geometry-aware metrics for pseudo-tracking diagnosis, (iii) energy-aware PPO fine-tuning for vertical and loop-like maneuver capability expansion, and (iv) executable target streams with closed-loop stream selection for long-horizon fixed-wing maneuver composition.

2 Related Work

Learning-Control Interfaces for Agile Flight. CoRL flight systems often place learning at an inspectable interface rather than replacing the full autonomy stack. Deep Drone Racing maps images to waypoints and desired speeds for downstream planning and control [1], and learning-based visual navigation combines learned perception with model-based planning and feedback [2]. Our interface is different: for fixed-wing maneuvering, the executable object is an attitude–speed target stream rather than a waypoint-speed command.

Direct Learned Flight Control. Direct learned controllers have been used for swarm flight and soft multicopter control [3, 4]. For fixed-wing aircraft, PPO-based attitude controllers have been studied in simulation and field experiments [5, 6]. Our policy is also direct at the actuator level, but remains target-conditioned rather than being an unconstrained end-to-end maneuver policy.

Fixed-Wing Guidance and Feasibility-Aware Tracking. Classical fixed-wing autonomy often separates path geometry, guidance, and inner-loop control. Dubins-style paths, nonlinear guidance laws, and vector-field methods provide standard tools for curvature-constrained fixed-wing path following [7, 8, 9, 10]. For agile robots, trajectory generation and receding-horizon methods reason about dynamic feasibility and finite-horizon objectives [11, 12, 13, 14], while DATT and learned time-allocation methods address infeasible or actuator-limited quadrotor references [15, 16]. In contrast, our target-stream selection checks feasibility through closed-loop execution by a learned fixed-wing executor.

Simulation and Evaluation. Fast simulation is central to robot learning, from Flightmare for quadrotors [17] to general or vehicle simulators such as MuJoCo, AirSim, FlightGoggles, and JSB-Sim [18, 19, 20, 21], as well as accelerator-oriented engines such as Brax and Isaac Gym [22, 23]. Our simulator is not the contribution of this paper; the evaluation gap addressed here is fixed-wing-specific: position error can hide invalid nose direction, velocity-tangent alignment, wing-plane orientation, energy state, angle of attack, sideslip, G-load, and termination risk.

Energy-Aware Target Streams for Learned Fixed-Wing Maneuvering

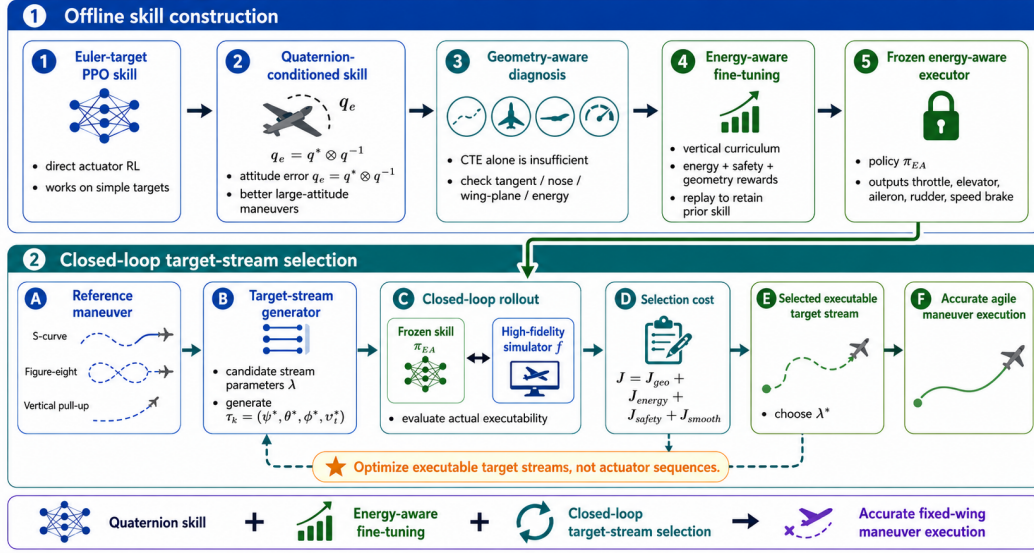


Figure 1: **System overview.** A direct-actuator fixed-wing skill is trained from Euler targets to a quaternion-conditioned, energy-aware executor. Long-horizon maneuvers are composed by converting references into target streams and selecting stream parameters through closed-loop rollout.

Algorithm 1 Energy-Aware Target Streams

Require: Dynamics f , maneuver set $\mathcal{D}$, stream generator G , candidates Λ

Ensure: Frozen skill π_{θ}^{EA} and executable rollouts

Offline skill construction

- 1: Train Euler-target skill: $\pi_{\theta}^{\text{Eul}} \leftarrow \text{PPO}(f, o^{\text{Eul}}, r_{\text{base}})$.
- 2: Train quaternion skill with $q_e = q^* \otimes q^{-1}$: $\pi_{\theta}^{\text{Quat}} \leftarrow \text{PPO}(f, o^{\text{Quat}}(q_e), r_{\text{base}})$.
- 3: Diagnose pseudo-tracking with $\mathcal{E}_{\text{geom}} = \{e_v, e_n, e_{nv}, e_w\}$, energy/safety metrics, and terminations.
- 4: Fine-tune: $\pi_{\theta}^{\text{EA}} \leftarrow \text{PPOFineTune}(\pi_{\theta}^{\text{Quat}}, r_{\text{base}} + r_{\text{energy}} + r_{\text{safety}} + r_{\text{geo}} + r_{\text{replay}})$.
- 5: Freeze π_{θ}^{EA} .

Closed-loop target-stream selection

- 6: **for** each maneuver $\mathcal{M} \in \mathcal{D}$ **do**
 - 7: **for** each $\lambda_i \in \Lambda$ **do**
 - 8: Generate $\tau_{0:T}^{(i)} = G(\mathcal{M}, x_0, \lambda_i)$, where $\tau_k^{(i)} = (\psi_k^*, \theta_k^*, \phi_k^*, v_{t,k}^*)$.
 - 9: Roll out $u_k^{(i)} = \pi_{\theta}^{\text{EA}}(x_k^{(i)}, \tau_k^{(i)})$, $x_{k+1}^{(i)} = f(x_k^{(i)}, u_k^{(i)})$.
 - 10: Score $J^{(i)} = w_g J_{\text{geo}}^{(i)} + w_e J_{\text{energy}}^{(i)} + w_s J_{\text{safety}}^{(i)} + w_u J_{\text{smooth}}^{(i)}$.
 - 11: **end for**
 - 12: Select $\lambda_{\mathcal{M}}^* = \arg \min_{\lambda_i \in \Lambda} J^{(i)}$ and execute the selected target stream.
 - 13: **end for**
-

3 Method

The system separates the reusable flight skill from the long-horizon composition layer. The skill provides a local attitude-speed tracking interface under actuator and aerodynamic constraints. The composition layer converts reference geometry into target streams and selects stream parameters by closed-loop rollout through the frozen skill. Algorithm 1 summarizes the complete pipeline.

3.1 Direct-Actuator Skill and Target Representation

The offline skill-construction stage in Algorithm 1 begins with direct-actuator RL and then replaces Euler target errors with quaternion attitude errors. The low-level policy is a recurrent GRU Actor-Critic trained with PPO [24], following standard advantage-estimation and on-policy implementa-

tion practices [25, 26]. It receives local attitude–speed targets and aircraft state features, and directly outputs throttle, elevator, aileron, rudder, and speed brake. Unlike an autopilot wrapper, this direct interface exposes actuator saturation, aerodynamic coupling, energy loss, and safety termination to the learned skill.

To handle large-attitude commands and overcome the mathematical singularities inherent to Euler angles, we use a quaternion-conditioned target representation. Specifically, the Euler representation suffers from gimbal lock as the pitch angle approaches $\pm 90^\circ$, losing a degree of freedom and causing erratic attitude errors during vertical pull-ups and loop-like maneuvers. By contrast, a quaternion formulation provides a singularity-free, continuous tracking error metric, consistent with common SO(3)-based attitude-control and estimation formulations [27, 28]. Let q be the current aircraft attitude and q^* the target attitude constructed from $(\psi^*, \theta^*, \phi^*)$. The quaternion-conditioned skill uses the shortest-rotation error in Eq. (1).

$$q_e = q^* \otimes q^{-1}, \quad q_e \leftarrow \text{sign}(q_{e,w})q_e, \quad d_{\text{SO}(3)}(q, q^*) = 2 \arccos |\langle q, q^* \rangle|. \quad (1)$$

The policy also observes body-frame target direction, speed/altitude features, angular rates, aerodynamic angles, and previous action. Euler-angle targets remain useful on simple commands, but Table 1 shows that quaternion conditioning is a better reusable executor for complex curve maneuvers.

3.2 Geometry- and Energy-Aware Skill Expansion

The offline skill-construction stage in Algorithm 1 uses geometry-aware metrics to diagnose pseudo-tracking before energy-aware fine-tuning. CTE alone cannot certify a valid fixed-wing maneuver. Given reference tangent $\hat{t}_{\text{ref}}$, velocity direction $\hat{v}$, body nose direction $\hat{n}_{\text{body}}$, body wing-plane direction $\hat{w}_{\text{body}}$, and reference plane normal $\hat{n}_{\text{plane}}$, we evaluate

$$\begin{aligned} e_v &= \arccos(\hat{v} \cdot \hat{t}_{\text{ref}}), & e_n &= \arccos(\hat{n}_{\text{body}} \cdot \hat{t}_{\text{ref}}), \\ e_{nv} &= \arccos(\hat{n}_{\text{body}} \cdot \hat{v}), & e_w &= \arccos(|\hat{w}_{\text{body}} \cdot \hat{n}_{\text{plane}}|). \end{aligned} \quad (2)$$

Together with speed, energy proxy, alpha/beta, G-load, action saturation, and termination reason, the metrics in Eq. (2) expose *pseudo-tracking*: low CTE with invalid attitude, energy, or safety state.

Starting from the quaternion skill, we fine-tune with an energy-aware vertical curriculum that mixes original targets, horizontal proxy tasks, level-altitude retention, pull-up tasks, and vertical arcs. The reward adds energy retention, low-speed safety, vertical-progress shaping, alpha/beta/G penalties, action smoothness, and geometry-alignment terms. Replay reduces forgetting of horizontal and mild-3D behavior. The fine-tuned skill is frozen before all target-stream experiments.

3.3 Executable Target Streams and Closed-Loop Selection

The target-stream selection stage in Algorithm 1 scores candidate target streams through closed-loop rollout. A reference maneuver $\mathcal{M}$ is not converted directly into actuator commands. Instead, a generator G maps the initial state and stream parameters λ to local attitude–speed commands,

$$\begin{aligned} \tau_{0:T} &= G(\mathcal{M}, x_0, \lambda), \\ \tau_k &= (\psi_k^*, \theta_k^*, \phi_k^*, v_{t,k}^*). \end{aligned} \quad (3)$$

Eq. (3) defines the executable target stream optimized by closed-loop selection. This interface is chosen instead of fixed waypoint tracking because waypoints specify where the aircraft should go, but not how attitude, lift direction, airspeed, and energy should evolve while reaching them. It is also chosen instead of full-horizon actuator optimization because actuator sequences are high-dimensional, maneuver-specific, and brittle under closed-loop policy errors. We therefore optimize low-dimensional stream parameters λ , which define an executable local contract between the reference geometry and the frozen learned skill.

We use deterministic lattice search because the current stream space is low-dimensional and feasibility is tested by closed-loop execution through the frozen policy. This gives grid-optimality within

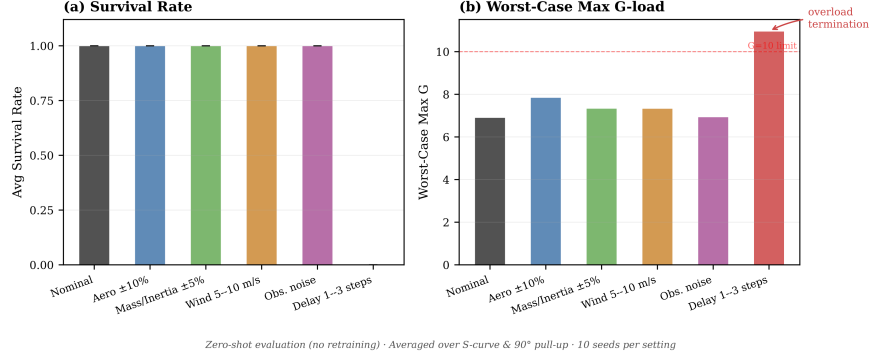


Figure 2: **Zero-shot perturbation sensitivity.** Survival remains high under aerodynamic, inertial, wind, and observation perturbations, but one-step actuator delay causes overload termination. Results average S-curve and 90° pull-up over 10 seeds.

the candidate set, not global optimality. We select stream parameters by closed-loop rollout through the frozen policy:

$$\begin{aligned} \lambda_{\mathcal{M}}^* &= \arg \min_{\lambda \in \Lambda} J(x_{0:T}^\lambda, u_{0:T-1}^\lambda, \mathcal{M}), \\ u_k^\lambda &= \pi_\theta(x_k^\lambda, \tau_k^\lambda), \quad x_{k+1}^\lambda = f(x_k^\lambda, u_k^\lambda). \end{aligned} \quad (4)$$

The selection in Eq. (4) scores target streams through closed-loop rollout, not open-loop geometry alone. The cost combines reference geometry, energy, safety, and smoothness:

$$\begin{aligned} J &= w_g J_{\text{geo}} + w_e J_{\text{energy}} \\ &\quad + w_s J_{\text{safety}} + w_u J_{\text{smooth}}. \end{aligned} \quad (5)$$

4 Experiments

4.1 Simulator and Evaluation Protocol

All experiments use Aeroplanax, a massively parallel fixed-wing RL benchmark with nonlinear 6-DOF dynamics, aerodynamic lookup tables, actuator limits, and vectorized JAX rollouts [29, 30, 31].¹ The simulator evaluates learned maneuvering under coupled attitude, lift, velocity, energy, and actuator constraints, with tabulated aerodynamic forces and moments, bounded aerodynamic angles, flight-envelope checks, low-speed/low-altitude terminations, load-factor limits, and out-of-bound conditions. Thus, success requires executable closed-loop flight under nonlinear aerodynamic and safety constraints, not merely low geometric tracking error.

This work is a simulation-stage study rather than a real-flight deployment claim. To probe transfer sensitivity, we evaluate the frozen executor without retraining under aerodynamic-coefficient scaling, mass/inertia scaling, wind disturbance, observation noise, and actuator delay on S-curve and 90° pull-up tasks with 10 seeds per setting. As summarized in Fig. 2, the tested aerodynamic, inertial, wind, and observation perturbations remain survivable, whereas one-step actuator delay causes overload termination.

4.2 Euler-to-Quaternion Skill Evolution

Table 1 compares Euler-angle and quaternion target encodings across maneuver regimes. The Euler baseline remains competitive on simple targets, but quaternion encoding substantially improves complex horizontal maneuvers such as circles, S-curves, and figure-eight trajectories. For example, on the right R5000 circle, attitude error decreases from 41.8° to 10.8°, and velocity-tangent error

¹An anonymized implementation of the simulation substrate is available at <https://anonymous.4open.science/r/Aeroplanax-2087/README.md>.

Table 1: Euler-angle and quaternion target encodings. Quaternion targets improve complex curve maneuvers, while Euler targets remain competitive on shallow loop-plane arcs.

Task	Att. Err. ($^{\circ}$)		Vel.-tan. ($^{\circ}$)		Wing-plane ($^{\circ}$)	
	Euler	Quat.	Euler	Quat.	Euler	Quat.
<i>Horizontal maneuvers</i>						
Circle $R5000$ right	41.8	10.8	33.4	5.8	41.1	10.6
Circle $R5000$ left	58.5	13.1	62.8	23.3	55.2	12.9
S-curve $A3000$	12.0	7.8	11.7	5.5	11.5	7.5
Figure-eight $R5000$	20.7	12.2	21.9	13.2	18.9	11.9
<i>Loop-plane arcs</i>						
60° loop arc	4.7	11.9	4.8	3.7	3.2	11.5

Table 2: Energy-aware vertical extension. The fine-tuned vertical skill preserves most original-task survival while correcting catastrophic failures on negative heading changes and reducing stall rates. Each task is evaluated with 10 seeds.

Task	Policy	Survival	Att. Err. ($^{\circ}$)	Stall Rate
Level flight	base skill	1.00	7.6	0.010
	fine-tuned vertical skill	1.00	7.3	0.002
Heading $+45^{\circ}$	base skill	1.00	12.8	0.000
	fine-tuned vertical skill	1.00	11.1	0.000
Heading -45°	base skill	0.00	56.9	0.000
	fine-tuned vertical skill	1.00	11.9	0.006
Heading -20°	base skill	0.00	62.2	0.375
	fine-tuned vertical skill	1.00	7.8	0.000
Circle $R5000$ right	base skill	1.00	11.0	0.008
	fine-tuned vertical skill	1.00	10.8	0.000
S-curve $A3000$	base skill	1.00	7.0	0.005
	fine-tuned vertical skill	1.00	7.8	0.000
30° pull-up $R8000$	base skill	1.00	17.7	0.024
	fine-tuned vertical skill	1.00	18.7	0.006

decreases from 33.4° to 5.8° . We therefore use the quaternion-conditioned policy as the reusable executor for target-stream composition, while avoiding the claim that it dominates Euler encoding on every target.

4.3 Pseudo-Tracking and Energy-Aware Extension

Fig. 3 (top) illustrates why CTE alone is insufficient. A trajectory can remain close to the spatial reference while exhibiting *flight-dynamics-inconsistent tracking*: poor velocity-tangent alignment, unsafe angle of attack, or degraded energy state. This motivates a *flight-dynamics-aware evaluation* that considers not only spatial error, but also tangent alignment, aerodynamic safety, energy state, and termination behavior.

Table 2 shows that energy-aware fine-tuning expands the safe and reliable envelope rather than uniformly improving every scalar metric. The fine-tuned skill preserves most original-task survival while correcting catastrophic failures on negative heading changes: heading -45° improves from survival 0.00 to 1.00, and heading -20° improves from survival 0.00 to 1.00 while stall rate drops from 0.375 to 0.000. It also reduces stall rates on retained tasks such as level flight, circle $R5000$ right, S-curve, and 30° pull-up, leading to more *flight-dynamics-consistent execution*.

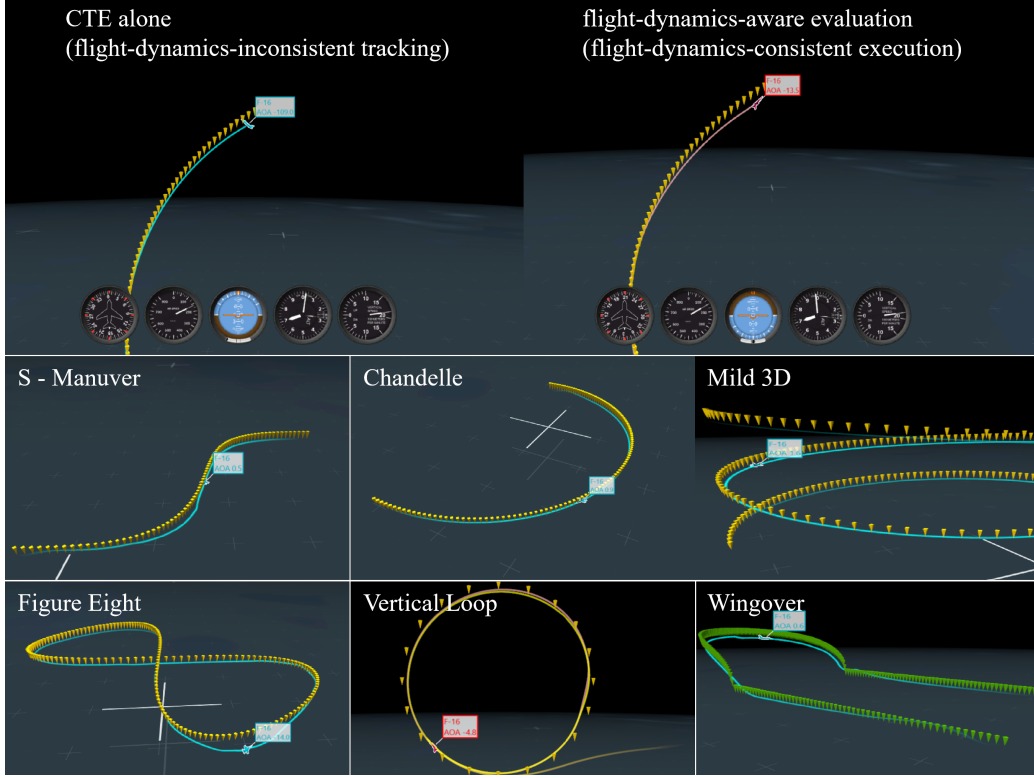


Figure 3: **Pseudo-tracking diagnosis and representative maneuvers.** *Top:* CTE-only tracking can remain spatially close while being flight-dynamics-inconsistent, whereas flight-dynamics-aware evaluation favors flight-dynamics-consistent, envelope-respecting execution. *Bottom:* The learned executor and selected target streams generate S-maneuver, chandelle, mild-3D, figure-eight, vertical-loop, and wingover-like maneuvers.

Table 3: **Target-stream selection and large-attitude maneuver evaluation.** Left: closed-loop target-stream sweep. Right: geometry-aware evaluation of vertical and loop-like maneuvers.

(a) Target-stream sweep				(b) Large-attitude evaluation				
Task	Best	CTE	Imp.	Arc	CTE	e_v	e_w	Grade
S-curve	(600, 220)	918/1503	39%	60°	180.0	12.8	103.6	B
Fig.-8	(600, 220)	1017/1039	2%	90°	152.5	10.1	66.2	B
Helix	(600, 220)	524/691	24%	105°	138.3	9.5	80.0	B
90° PU	(1500, 280)	51/96	47%	120°	121.3	8.8	69.5	B
CTE is best/default; default stream is (1000, 250).				135°	120.1	6.9	60.4	B
				150°	181.2	6.6	49.2	B
				Fullloop	181.0	6.7	45.1	B

4.4 Target-Stream Selection and Large-Attitude Maneuvers

Table 3(a) reports the closed-loop stream sweep. Horizontal and mild-3D curves prefer shorter lookahead and lower speed, while the vertical pull-up prefers longer lookahead and higher speed, supporting task-dependent stream selection over a single fixed parameter. The selected streams improve CTE-P90 by 39%, 2%, 24%, and 47% on S-curve, figure-eight, helix, and 90° vertical pull-up tasks.

Fig. 3 (bottom) shows executable maneuvers generated by the proposed skill and stream interface, not benchmark-provided demonstrations. Table 3(b) further shows that, after energy-aware fine-tuning, the frozen PPO executor completes vertical-arc maneuvers and can be composed into a full loop-like maneuver. The remaining errors are therefore better interpreted as execution-quality and energy-margin limitations rather than basic maneuver-completion failures.

183 4.5 Comparison with Classical Cascade Autopilot

184 We compare the learned executor with a classical cascade autopilot, using nonlinear guidance plus
 185 a tuned PID inner loop, on the 90° vertical pull-up. We tested both a 50 Hz inner / 5 Hz outer
 186 multi-rate setup and a synchronous 5 Hz setup, and report the more stable synchronous baseline to
 187 avoid a strawman comparison. As shown in Fig. 4, the PID+NLGL baseline initially achieves lower
 188 CTE-P90 than the selected target stream (113 m vs. 170 m), but this geometric aggressiveness drives
 189 pitch commands above 45° , AoA above 18° , and load factor to 9.6 g, causing overload termination
 190 at 6,985 m. The learned executor accepts slightly larger CTE while maintaining AoA below 4° ,
 191 load factor below 7 g, and airspeed around 230 m/s, completing the 14.7 km climb. This contrast
 192 illustrates pseudo-tracking: minimizing spatial error alone can produce dynamically invalid fixed-
 193 wing maneuvers.

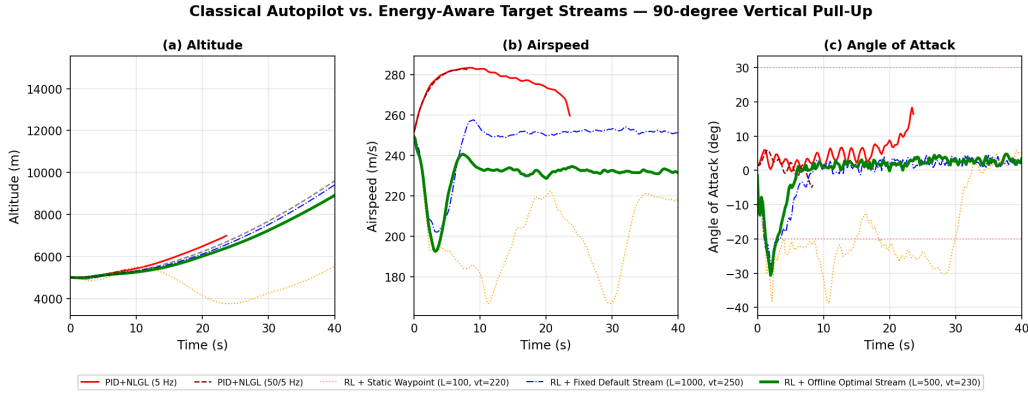


Figure 4: **Classical autopilot vs. learned executor.** On a 90° pull-up, PID+NLGL reduces early CTE but terminates from energy/load-factor failure; the learned executor preserves energy and completes the climb.

194 5 Discussion and Limitations

195 The learned skill is a direct-actuator executor with an explicit attitude–speed interface, not a full
 196 autonomy stack. Closed-loop stream selection searches low-dimensional target-stream paramet-
 197 ers rather than actuator sequences; the current deterministic lattice is only grid-optimal within
 198 the enumerated candidates and could be replaced by sampling-based optimizers such as CEM or
 199 MPPI [14, 13].

200 This remains a simulation-stage result. Despite nonlinear 6-DOF dynamics, aerodynamic lookup
 201 tables, actuator/load constraints, and safety terminations, real-flight transfer still requires latency-
 202 aware actuation, broader domain randomization, safety monitoring, and progressive validation. The
 203 perturbation study identifies actuator delay as a critical failure mode; relevant sim-to-real directions
 204 include domain randomization [32, 33], iterative adaptation [34], and robust policy transfer [35].

205 6 Conclusion

206 We presented energy-aware target streams for learned fixed-wing maneuvering. The framework
 207 combines a quaternion-conditioned direct-actuator skill, geometry-aware pseudo-tracking diagno-
 208 sis, energy-aware capability expansion, and closed-loop executable target-stream selection. Experi-
 209 ments show that quaternion targets improve complex curve maneuvers, energy-aware fine-tuning im-
 210 proves vertical-task safety and reliability, and task-dependent target streams reduce CTE-P90 across
 211 horizontal, mild-3D, and vertical pull-up maneuvers. The fine-tuned executor can further compose
 212 selected target streams into loop-like large-attitude maneuvers in simulation, while geometry-aware
 213 metrics reveal remaining limitations in velocity alignment, wing-plane consistency, safety margins,
 214 and energy management.

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